Lesson 5 – Probability and Statistics Notations

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October 13, 2026

Basic Probability

Probability of an event A occuring is written as $P(A)$, where:

$$0 \leq P(A) \leq 1$$

Conditional Probability

The conditional probability of event A given B is:

$$P(A \mid B)$$

This is read as *" The probability of A given B."*

Intersection and Union

The probability of both event A and B occuring simultaneously is:

$$P(A \cap B)$$

The probability of either event A or B occuring is:

$$P(A \cup B)$$

Compliment

The compliment of an event A is the probability that A does not occur:

$$p(A^c)$$

This follows the fundamental rule:

$$P(A) + P(A^c) = 1$$